

Machine Learning Fundamentals - Lecture Notes (Week 3)

Gradient Descent

Gradient descent minimises a loss function by repeatedly moving the parameters in the direction of the negative gradient. The update rule is: new weights = old weights minus the learning rate times the gradient of the loss.

The learning rate controls the step size. If it is too large the loss can oscillate or diverge; if it is too small training is very slow. Stochastic gradient descent estimates the gradient from a single example or a small mini-batch instead of the full dataset, which makes each step cheaper but noisier.

Momentum keeps a running average of past gradients, which helps the optimiser move faster along consistent directions and dampens oscillations.

Overfitting and Regularisation

A model overfits when it learns noise in the training data and performs much worse on unseen data. Typical signs are a training loss that keeps falling while the validation loss starts to rise.

Remedies:

- Collect more training data.
- L2 regularisation (weight decay) adds the sum of squared weights to the loss, which discourages large weights.
- L1 regularisation adds the sum of absolute weights and tends to produce sparse models where many weights are exactly zero.
- Dropout randomly disables a fraction of neurons during training so the network cannot rely on any single unit.
- Early stopping ends training when the validation loss stops improving.

Evaluating Classifiers

Accuracy is the fraction of correct predictions. It can be misleading on imbalanced datasets: if 95 percent of emails are not spam, a model that never predicts spam is 95 percent accurate.

Precision is the fraction of predicted positives that are actually positive: $TP / (TP + FP)$. Recall is the fraction of actual positives that the model finds: $TP / (TP + FN)$. The F1 score is the harmonic mean of precision and recall.

K-fold cross-validation splits the data into k parts, trains on k-1 parts and tests on the remaining part, repeating k times. The average score is a more reliable estimate than a single train/test split, especially for small datasets.